



STANDALONE SMART HAJJ EARBUD ASSISTANT

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How Can Technology Improve Pilgrim Safety During Hajj?

Hajj is one of the largest religious gatherings in the world, attracting millions of Muslims each year. Despite its religious importance, pilgrims face challenges such as extreme temperatures, overcrowding, long walking distances, and language barriers, which can affect their health and safety.

The device aims to improve communication, enhance pilgrim safety, and support the goals of Saudi Vision 2030 in providing a safer and smarter Hajj experience.

Overview & Problem Background

- Hajj faces major challenges: extreme heat, overcrowding, and language barriers.
- These factors increase health risks and delay emergency response.
- The smart earbud provides vital-sign monitoring, translation, emergency alerts, and direct communication with medical teams.
- Goal: improve pilgrim safety and reduce preventable health complications.

Problem Statement

- Pilgrims face health risks from heat, congestion, and communication difficulties.
- Existing systems lack continuous monitoring and fast multilingual support.
- Delays in detecting health problems can increase emergencies.
- A smart wearable is needed to monitor vital signs and assist communication in real time.

Objectives

- Monitor heart rate, temperature, and blood oxygen levels.
- Provide multilingual translation.
- Identify pilgrim location using GPS.
- Send emergency SOS alerts.
- Enhance a safe and smart Hajj experience.
- Support medical teams with accurate health and location data.

Contributions / Significance

- Enhances pilgrims' health and safety through continuous vital-sign monitoring.
- Improves communication through multilingual translation.
- Enables rapid emergency response with real-time alerts.
- Provides an innovative smart solution for serving pilgrims.

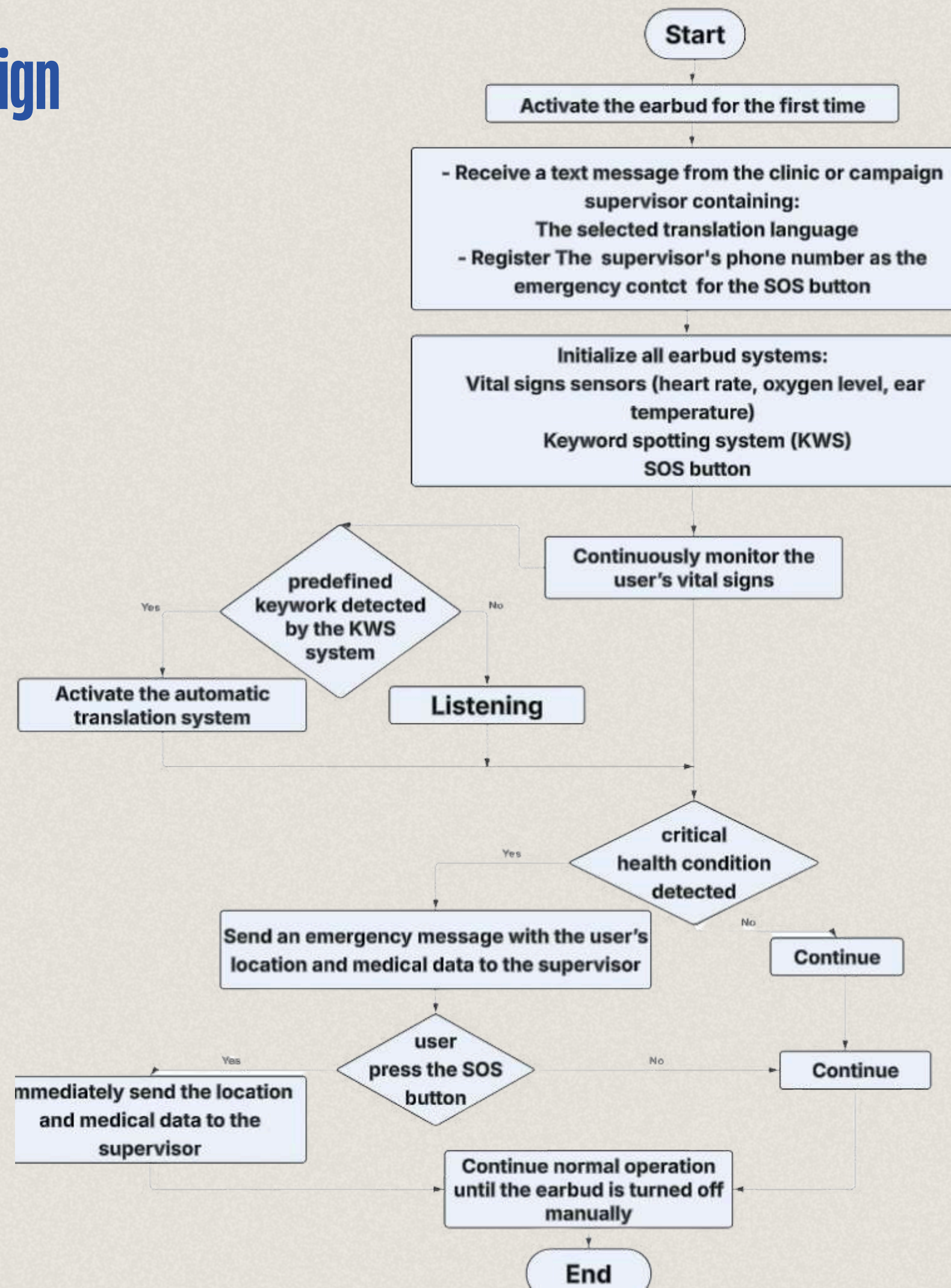
Realistic Constraints

- Prioritizes pilgrims' health and safety.
- Ensures ethical use for medical and humanitarian purposes.
- Designed for sustainability with energy-efficient hardware.
- Supports communication between pilgrims and medical teams.
- Uses locally available and reliable components.
- Environmentally safe and suitable for Hajj conditions.



System Analysis and Design

Flowchart



Hardware Specifications

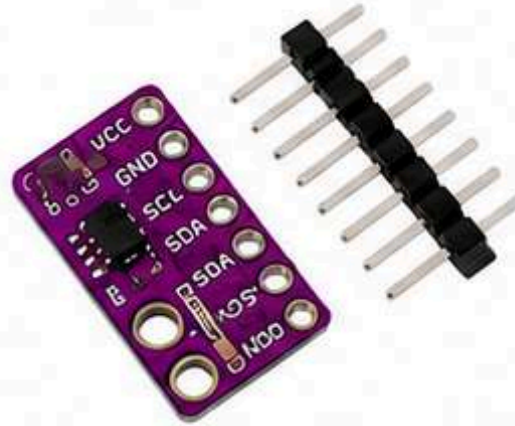
**ESP32-S3-DevKitC-1-
N32R16V**



PPG sensors



MAX30205



SIM7600E



IN

MP441



SIM card + Holder



3.7V 500-800mAh Li-po



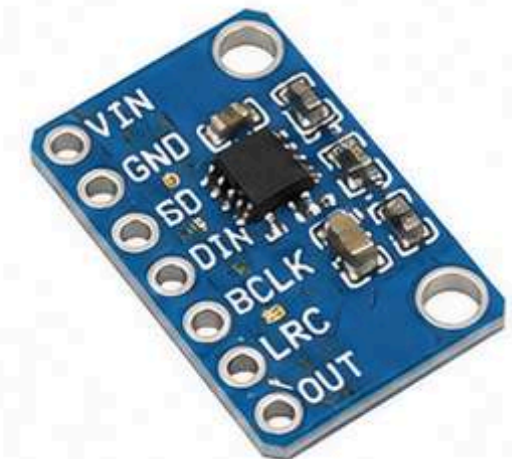
**Tactile push
button SOS**



Mini speaker



**MAX98357A
I2S Audio Amplifier**

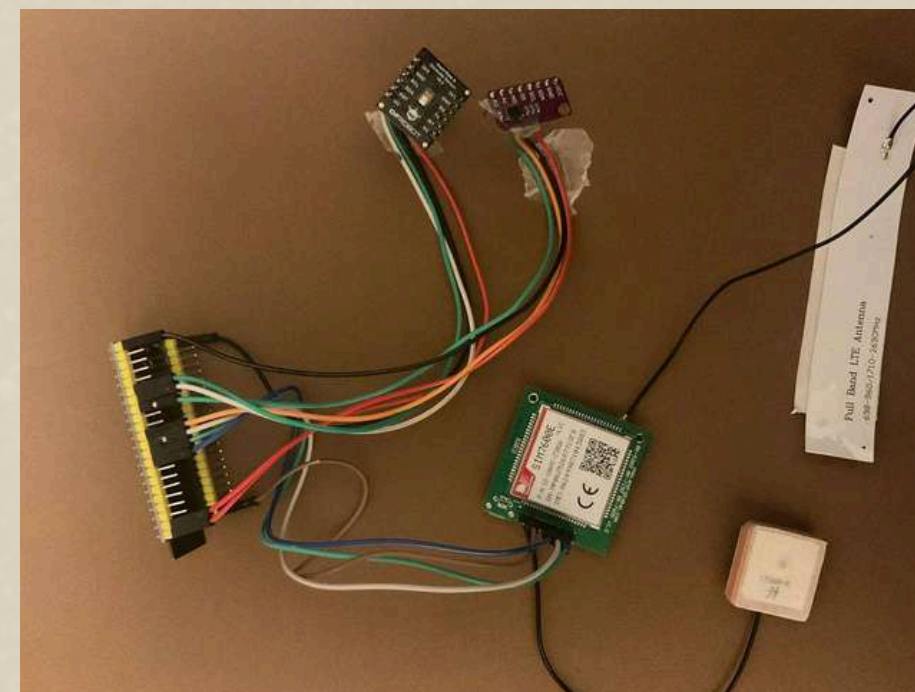
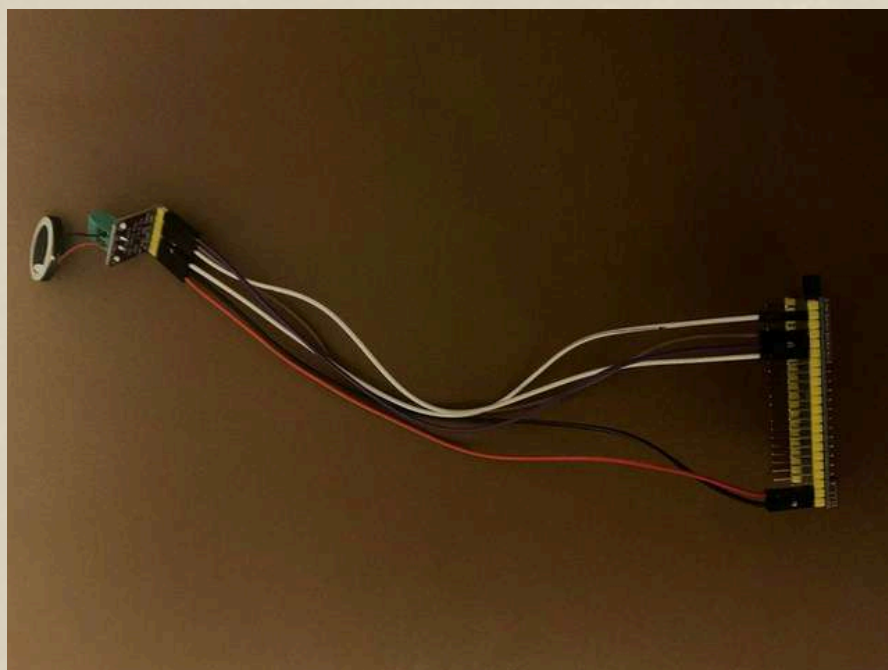




Wire Harnessing and Cable Management

Contributions / Significance

- EnSignal Segregation: Analog, digital, and RF signals are separated into distinct bundles to eliminate electromagnetic interference (EMI).
- Audio Sub-Harness (I²S): MAX98357A amplifier is directly wired to the 8 ohm 1W speaker with minimal wire length to prevent audio distortion.
- Sensors Harness (I²C): PPG and MAX30205 temperature sensors share a dedicated, low-noise I²C bus path.
- RF Shielding: The SIM7600E LTE module and antenna are isolated on separate serial lines to protect medical sensors from transmission noise.
- Color-Coding: Utilized color-coded jumper wires to optimize the layout and simplify hardware debugging and maintenance.



The system was developed by testing each hardware and communication module independently to ensure accurate and reliable performance. After successful validation, all modules were integrated into a complete system to verify seamless operation and efficient interaction between hardware and software components.

PPG Sensor (heart rate and blood oxygen)

- PPG Sensor (Heart Rate & blood oxygen): Tested on multiple individuals and compared with global medical baselines, The obtained values showed high accuracy and were within the acceptable medical range:
 - Normal Heart Rate Range: 60 - 100 BPM
 - Normal Blood Oxygen Saturation: 95% - 100%

```
Stabilizing... Please hold your finger still.  
Stabilizing... Please hold your finger still.  
SPO2: 98%  
Heart Rate: 73 bpm  
-----
```

```
37 void loop() {  
38     MAX30102.getHeartbeatSPO2();  
39  
40     int rawSPO2 = MAX30102._sHeartbeatSPO2.SPO2;  
41     int rawHeartRate = MAX30102._sHeartbeatSPO2.Heartbeat;  
42  
43  
44     if (rawHeartRate > 45 && rawHeartRate < 150) {  
45  
46         heartRateBuffer[filterIndex] = rawHeartRate;  
47         filterIndex = (filterIndex + 1) % FILTER_SIZE;  
48  
49  
50  
51         long sum = 0;  
52         for (int i = 0; i < FILTER_SIZE; i++) {  
53             sum += heartRateBuffer[i];  
54         }  
55         int filteredHeartRate = sum / FILTER_SIZE;  
56  
57  
58         Serial.print("SPO2: ");  
59         Serial.print(rawSPO2);  
60         Serial.println("%");  
61  
62         Serial.print("Heart Rate: ");  
63         Serial.print(filteredHeartRate);  
64         Serial.println(" bpm");  
65  
66  
67     } else {  
68         Serial.println("Stabilizing... Please hold your finger still.");  
69     }  
70  
71     delay(1000);  
72 }
```


MAX30205 temperature sensor

- comprehensively tested to verify its functionality in monitoring body temperature. The readings were compared against the normal human baseline, which is considered to be around 36.5–37.5°C.

--- نظام قياس الحرارة ---
36.77 °C درجة حرارة الجسم التقريبية :
36.77 °C درجة حرارة الجسم التقريبية :

```
14 void loop() {  
15     Wire.beginTransmission(LM75_ADDR);  
16     Wire.write(0x00);  
17     Wire.endTransmission();  
18  
19     Wire.requestFrom(LM75_ADDR, 2);  
20  
21     if (Wire.available() >= 2) {  
22         int8_t msb = Wire.read();  
23         uint8_t lsb = Wire.read();  
24  
25  
26         int16_t combined = (msb << 8) | lsb;  
27         float raw_temperature = combined / 256.0;  
28  
29  
30         float calibrated_temperature = raw_temperature + THERMAL_OFFSET;  
31  
32  
33         Serial.print("درجة حرارة الجسم التقريبية :");  
34         Serial.print(calibrated_temperature);  
35         Serial.println(" °C");  
36  
37  
38     } else {  
39         Serial.println("خطأ: تعذر الاتصال بالمحساس، تأكد من ثبات الأسلاك");  
40     }  
41  
42     delay(5000); // تحديث كل 5 ثوانٍ
```


4G & GPS Testing

- Wireless communication and location tracking modules were rigorously tested under various scenarios to ensure maximum reliability.
- Data Transmission: The system successfully established a network connection, transmitting GPS coordinates and vital signs via SMS

```
[START] SMS... النظام العطور لإرسال إلى
[POWER] ...جاري إرسال تهيئة التشغيل للمودم
[WAIT] ...الانتظار لمدة 30 ثانية لتحميل الشريحة واستقرار الجهد
30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1
[INFO] (Serial2) ... انتهى وقت الانتظار. جاري فتح السيريال العنابي الثاني
[CHECK] (AT) ... إرسال أمر فحص الاتصال التلقائي

[SMS] (Text Mode) ... تحويل المودم إلى وضع النصوص 1.

[SMS] ...إرسال الرقم المستهدف 2.

[SMS] ...إرسال نص الرسالة 3.
[SMS] (Ctrl+Z) ... إرسال رمز إنهاء الرسالة وأمر الإرسال الفعلي 4.
[SMS] ... (المنظري 10 ثوان) Message Reference جاري انتظار رد الشبكة النهائي وال 5.

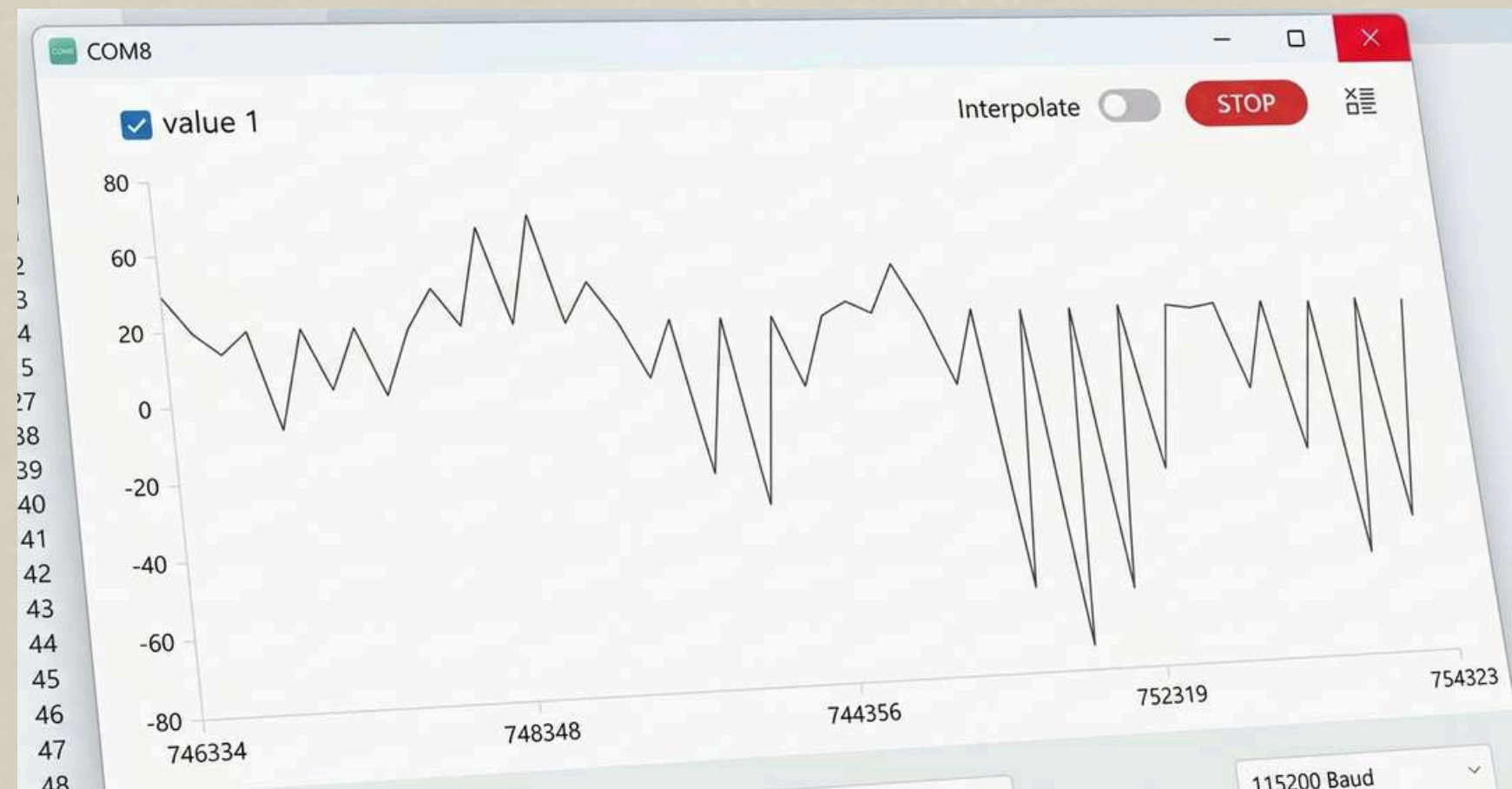
[FINISH] ... انتهت المحاولة. راقبي الودود بالأعلى وجوانك
```

iMessage
Tuesday 8:53 pm

Hello! 4G SMS Test from ESP32-S3 and
SIM7600E is successful! 🎉

microphone and speaker modules

- Audio Efficiency: Evaluated microphone capture and calibrated the speaker for crystal-clear sound and pronunciation.
- Crowd Performance: Demonstrated smooth, low-latency audio operation, ensuring effective guidance in noisy environments.



Dataset & Pipeline

Collected 12,000 Arabic audio samples for 29 Hajj-related keywords and prepared them for training.

Mel-Spectrogram

Audio signals were converted into Mel-spectrograms to enable efficient feature extraction.

CNN Architecture

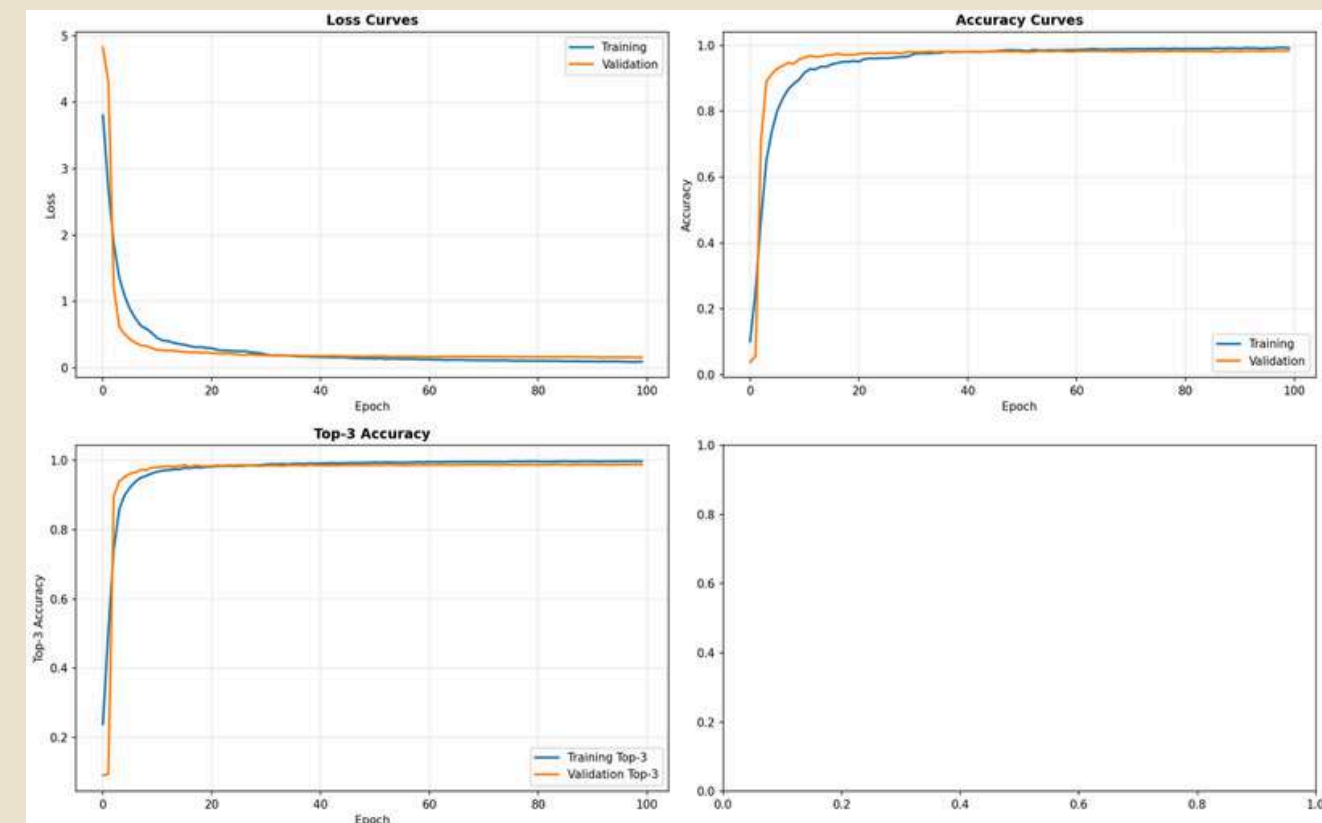
A custom CNN model was developed to classify 29 Arabic keywords with high accuracy.

Model: "sequential"

Layer (type)	Output Shape	Param #
normalization (Normalization)	(None, 63, 64, 1)	3
conv2d (Conv2D)	(None, 63, 64, 32)	320
batch_normalization (BatchNormalization)	(None, 63, 64, 32)	128
conv2d_1 (Conv2D)	(None, 63, 64, 32)	9,248
batch_normalization_1 (BatchNormalization)	(None, 63, 64, 32)	128
max_pooling2d (MaxPooling2D)	(None, 31, 32, 32)	0
spatial_dropout2d (SpatialDropout2D)	(None, 31, 32, 32)	0
conv2d_2 (Conv2D)	(None, 31, 32, 64)	18,496
batch_normalization_2 (BatchNormalization)	(None, 31, 32, 64)	256
conv2d_3 (Conv2D)	(None, 31, 32, 64)	36,928
batch_normalization_3 (BatchNormalization)	(None, 31, 32, 64)	256
max_pooling2d_1 (MaxPooling2D)	(None, 15, 16, 64)	0
spatial_dropout2d_1 (SpatialDropout2D)	(None, 15, 16, 64)	0
conv2d_4 (Conv2D)	(None, 15, 16, 128)	73,856
batch_normalization_4 (BatchNormalization)	(None, 15, 16, 128)	512
conv2d_5 (Conv2D)	(None, 15, 16, 128)	147,584
batch_normalization_5 (BatchNormalization)	(None, 15, 16, 128)	512
max_pooling2d_2 (MaxPooling2D)	(None, 7, 8, 128)	0
spatial_dropout2d_2 (SpatialDropout2D)	(None, 7, 8, 128)	0
global_average_pooling2d (GlobalAveragePooling2D)	(None, 128)	0

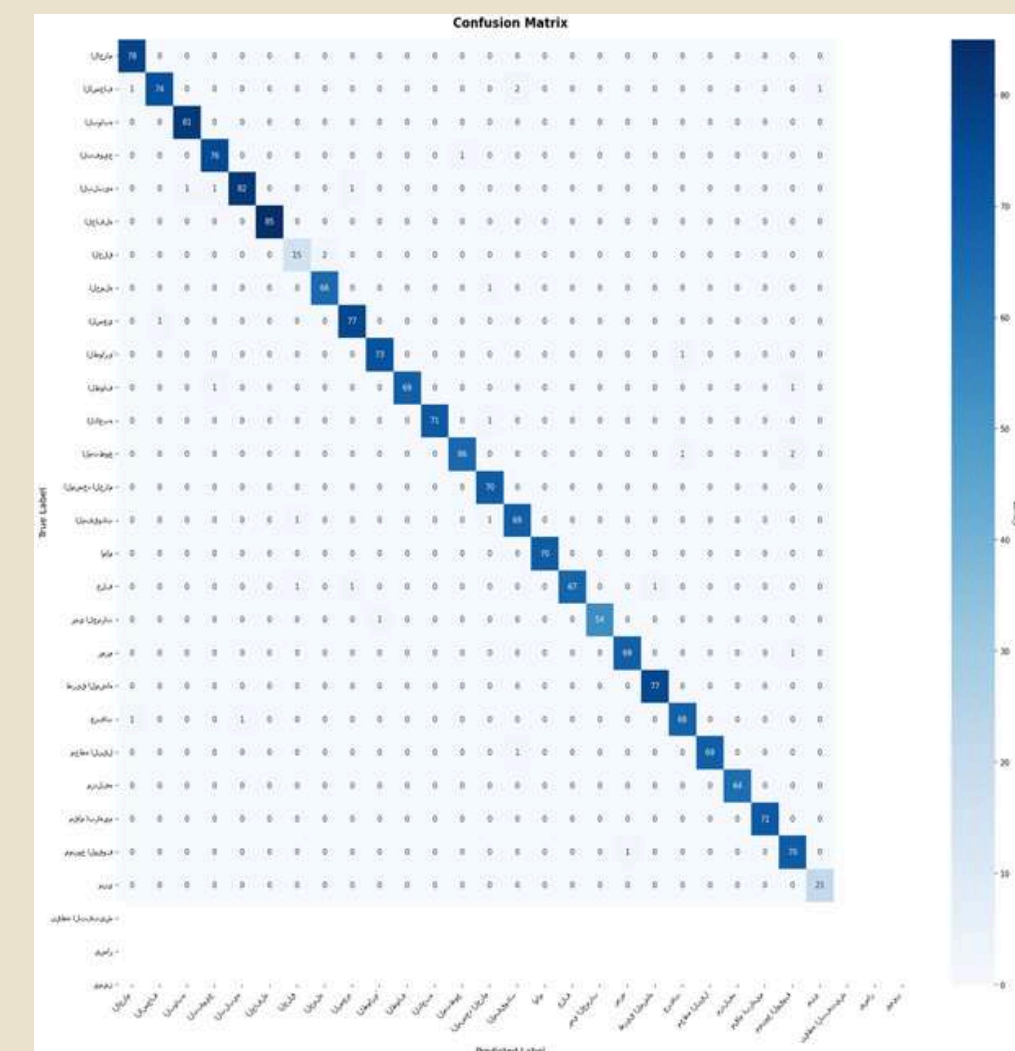
Training Results

The model achieved approximately 97–99% accuracy and demonstrated strong performance during testing.



Confusion Matrix

Results showed excellent classification with minimal confusion between similar words.



ESP32-S3 Deployment

The trained model was converted to TensorFlow Lite and successfully deployed on the ESP32-S3 microcontroller.

3D Design and Integrated Health Monitoring System

The Smart Hajj Earbud Assistant features a comfortable U-shaped neckband for secure wear and balanced weight distribution. It includes wired earbuds, power and SOS control buttons, and ventilation openings for enhanced comfort. Integrated sensors continuously monitor body temperature, heart rate, and blood oxygen saturation (SpO₂).

Front View



Side View



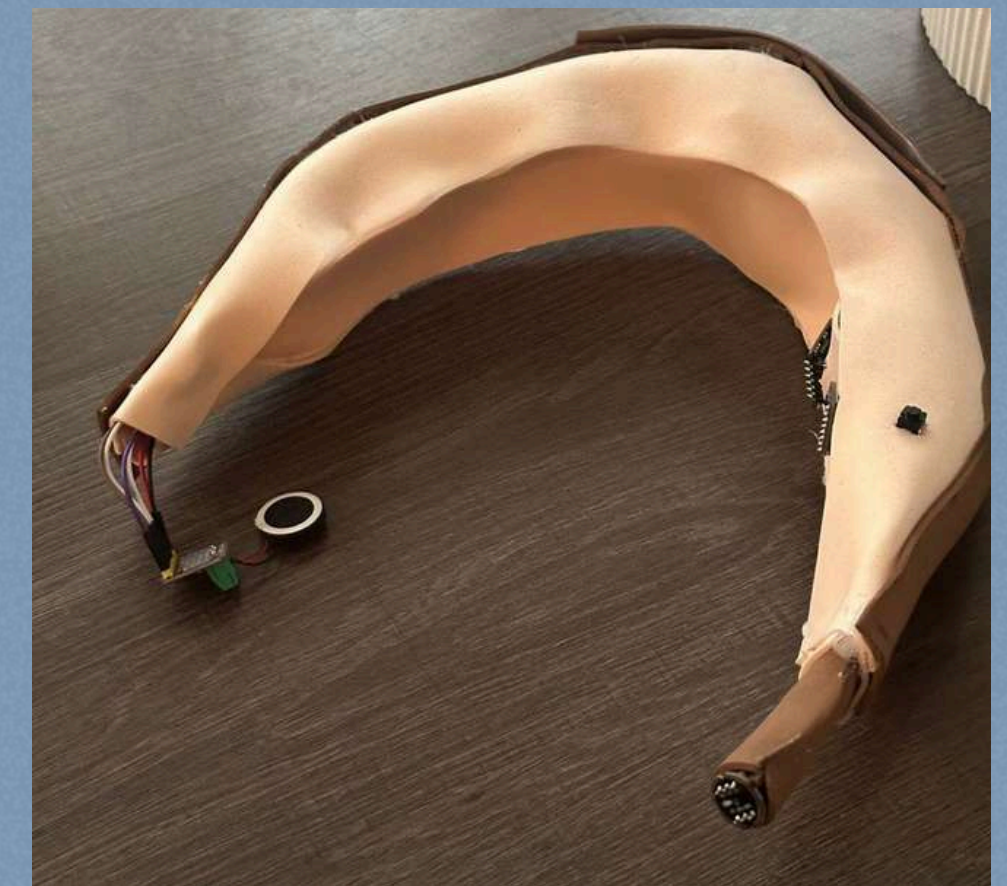
Back View



■ System Result And Testing

After integrating all hardware components, the system was implemented using the ESP32-S3 to manage sensor monitoring, translation, GPS tracking, and SOS alerts. Final testing confirmed stable performance, successful communication between components, and reliable transmission of the pilgrim's vital signs and real-time location through the 4G module.

The final results demonstrated reliable operation and successful communication between all hardware and software components.



The project presents a smart in-ear health monitoring system that combines vital signs monitoring, keyword spotting, translation, GPS tracking, and SOS emergency alerts. By utilizing the ear canal for accurate measurements, the system enhances pilgrim safety, communication, and healthcare support, contributing to a smarter Hajj experience.

The system was successfully implemented and tested with all hardware components integrated into a functional prototype. Future improvements may include enhancing the earbud design, optimizing battery performance, and conducting real-world testing during Hajj to further evaluate reliability and user experience.

THANK YOU

